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RESEARCH ARTICLE

Linear Programming Approach: Optimization of Profit and Land Resource

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ABSTRACT

Allocating resources like land, fertilizer, crops, water, etc. requires a detailed farm plan. In this study, the LINEAR PROGRAMMING MODEL is used to identify crop combinations for a farmer in the Gurugram district of Haryana. This work provides a Linear Programming Model for RABI crops (Wheat, Mustard, and Barley (Joo)) and for KHARIF crops (Millet(bajra), radish, and rice). The optimal allocation of land is obtained with the use of MATLAB, which improves the quality of decisions. It was found that the profit can be increased by 24% for the RABI crop and 6.7% can be increased for the KHARIF crop. Also, a graph showing the comparison of the current situation and the optimal solution is shown (Fig. no.2 &3). To create the perfect model, some farm attributes—such as climate and market fluctuations—are kept constant.

Keywords: Linear programming, current situation, Optimal solution, Attributes, Market fluctuations, etc

INTRODUCTION

The agriculture sector plays a significant role in the social and economic development of India. Meeting the growing demands for food, feed, and fibre presents ever-greater challenges for agriculture, the foundation of our global economy. This vital industry that provides the majority of the world's food, faces the difficult task of balancing productivity and resource use in the face of changing social, economic, and environmental factors. As the study done by Joshi, A., Das, S., & VA seem, Mohd. [2] States that India's agricultural industry has numerous obstacles resulting from variables relating to markets, policies, the environment, and structure. The modern period of changing climatic circumstances and population expansion has made it essential to allocate resources optimally and maximize output.

The optimal use of land and water resources is necessary for agricultural production. This means that a farm's resources must be distributed effectively. Choosing what to grow, when to plant it, what agricultural techniques are available, and how much is needed are tough decisions that farmers must make. The revolutionary potential of decision support systems in promoting sustainable development in the digital era and modernizing agricultural practices is highlighted by Zhai, Martínez, Beltran, and Martínez [10]. The financial and physical state of farms are taken into consideration when making decisions, but there is still a lot of uncertainty over this industry's planning horizon. Resources such as workers, seeds, manures, and fertilizers might have erratic production and pricing (Priya Priyadarshi and Purushothaman [8].

In order to make better decisions, allocate resources optimally, and increase the resilience, sustainability, and efficiency of agricultural systems, linear programming is essential. Its applications cover a wide range of management, policy, and production elements of agriculture and improve sustainable agriculture and rural development globally. This study sets out to investigate the use of linear programming, a mathematical method of optimization, in the field of agriculture. The work presented by N. A. Sofi, Aquil Ahmed, Mudasir Ahmad, and Bilal Ahmad Bhat [6]focuses on using linear programming (LP) as an approach to successfully handle these issues. Also, in a study of Senthilnathan S. [9] they highlighted the potential of linear programming models with the use of the MS Excel in optimizing crop selection and resource allocation in Central Greece. Based on mathematical modelling, linear programming offers a methodical way to make decisions. Majeke, F., Mubvuma, M. T., and Makaza, K. [3,4,5] worked to find the best crop combination for a rural farmer, a linear programming model was used for this study. The credit restriction was integrated into the linear programming paradigm. To maximize revenue was the goal. The objective of this work is to explore the various aspects of the linear programming approach in agriculture, clarifying its past development, present uses, and prospective future developments. Moreover, linear programming can assist in long-term planning and decision-making.

Long-term planning and decision-making can benefit from the use of linear programming. Farmers can predict future prices and demand by analysing market patterns and historical data. Alanoud and Farrukh [1]farmers can make proactive decisions on crop selection, planting schedules, and resource allocation by integrating these forecasts into the linear programming model. This will help them stay ahead of market fluctuations. The study by Poonia, Tonk, Bhatia, etc. all[7] provide a useful tool for improving farm profitability and sustainability in resource-constrained contexts by utilizing the power of linear programming.

We find the optimal solution understanding and utilizing the power of linear programming becomes essential for sustainable and effective agricultural practices in an era climate uncertainty and pressures from a growing global population need careful resource management. Therefore, this study's goals go beyond merely outlining its methods; they also include a sincere attempt to clarify the revolutionary role that linear programming can play in influencing how agriculture develops in the future. We find the optimal solution Understanding and utilizing the power of linear programming becomes essential for sustainable and effective agricultural practices in an era climate uncertainty and pressures from a growing global population need careful resource management. Therefore, this study's goals go beyond merely outlining its methods; they also include a sincere attempt to clarify the revolutionary role that linear programming can play in influencing how agriculture develops in the future.

An LP model was developed in this study to ascertain the best crop combination for a farmer in Gurugram district of Haryana, India. Wheat, Mustard and Barley (Joo) were the crops taken into account in Rabi season and Millet(bajra), radish and rice were the crops taken into account in Kharif season. The optimal solution to the problem is obtained using MATLAB. Our goal is to offer insightful information on the intricate dynamics involved in making agricultural decisions. Hence a comparison has also been done of the current situation and optimal solution suggested using LP Model with the help of MATLAB. This comparison is shown by the graph (Figure 2 and 3). The following diagram shows Optimization technique:

METHODOLOGY

A linear programming with ''n`` decision variables and ''m`` constraints can be mathematically modelled as:

$$Z_{max} = c_1x_1 + c_2x_2 + c_3x_3 + \cdots \ldots \ldots \ldots . c_nx_n$$
$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots \ldots \ldots \ldots . a_{1n}x_n \leq b_1$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots \ldots \ldots \ldots . a_{2n}x_n \leq b_2$$
$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots \ldots \ldots \ldots . a_{mn}x_n \leq b_m$$
$$x_{ij} \geq 0 , j = 1 ,2 \ldots \ldots \ldots \ldots n$$

Data Collection

The data is collected from a small village farmer of the Gurugram district of Haryana for both the seasons, Kharif and Rabi. The crops sown by the farmer in Rabi season the crops are Wheat, Mustard, and Barley (Joo) whereas in the kharif season are Millet (Bajra), Rice, and Radish. Data regarding land allocation, man-days, costs, and the net return of each crop required for each crop is based on recent methodology used by the farmer for both seasons. The LP problem is solved by using "MATLAB", a computer programming platform.

The LP (linear programming) model is developed using exact data given by farmer to maximize the net return at the end of the Kharif and Rabi seasons.

RESULTS AND DISCUSSION

The LP (linear programming) model is developed using exact data given by farmer to maximize the net return at the end of the Rabi and Kharif seasons.

RABI SEASON

Table **1** shows the farmer's cropping pattern (allocation of land, man-days, and operating capital required for the preparation of the crops). As the table shows the farmer uses a total of 40 acres of land for Wheat, Mustard, and Barley (Joo) with 500 total man-days and operating capital of rupees 550000 in the Rabi season which gives the net returns of 1666000 rupees. Also, as per the farmer's requirements, he needs at least 3 acres of land for wheat, at least 2 acres for mustard, and 2 acres of land for Barley.

In this scenario, the output of different crops serves as the objective function, while the land, man-days, and operating capitalrepresent the inequalities and the total requirement. Finding the ideal area for crops is now our goal.

LP model

$$Z_{max} = 50000x_1 + 52000x_2 + 48000x_3$$
$$S.to.\ x_1 + x_2 + x_3 \leq 40$$
$$18000x_1 + 14000x_2 + 15000x_3 \leq 750000$$
$$11x_1 + 4x_2 + 11x_3 \leq 500$$
$$x_1 \geq 3, x_2 \geq 2, x_3 \geq 2$$
$$x_i \geq 0 , i = 1 ,2,3$$

where x_1= acres of land allocated to Wheat
x_2 = acres of land allocated to Mustard
x_3= acres of land allocated to Barley

The LP model's recommended cropping pattern (Table 2) demonstrates that returns can be increased by allocating 3 acres of land to Wheat,35 acres to Mustard, and 2 acres of land to Barley. Hence according to the cropping pattern recommended by the LP model planting more Mustard on available land can increase returns. Since the land allocation in the LP model is different from land allocation by the farmer, so Man-days and Costs will also vary accordingly.

Thus, it is evident from the comparison graph and table that the cropping pattern suggested by the LP model can raise the net return by 24%.

KHARIF SEASON

The farmer planted Millet (Bajra), rice, and Radish as their crops. Table 3 shows that the average amount of resources available was 32 acres of land. The farmer leaves 8 acres of vacant land as fallow land,800 man-days, and 300000 rupees in operating capital during the Kharif season. Planning an appropriate farming pattern was intended to increase yields relative to the current net average return of farmers, which was rupees 1867800.Also, the farmer requires at least 10 acres of land for Millet (Bajra), at least 5 acres of land for Rice, and at least 5 acres of land for Radish.

LP Model

$$Z_{max} = 35000x_1 + 70400x_2 + 85000x_3$$
$$S.to.\ \ x_1 + x_2 + x_3 \leq 32$$
$$15000x_1 + 28000x_2 + 33400x_3 \leq 950000$$
$$15x_1 + 25x_2 + 40x_3 \leq 800$$
$$x_1 \geq 10, x_2 \geq 5, x_3 \geq 5$$
$$x_i \geq 0\ , i = 1\ ,2,3$$

where x_1=acres of land allocated to Millet (Bajra)
x_2=acres of land allocated to Rice
x_3=acres of land allocated to Radish

The LP model's recommended cropping pattern (Table 5) demonstrates that returns can be increased by allocating 10 acres of land to Millet (Bajra) ,15.34 acres to Rice and 6.67acres of land to Radish. Since the land allocation in LP model is different from land allocation by farmer, so Man-days and Costs will also vary accordingly**.** Thus, it is evident from the comparison graph and table that the cropping pattern suggested by the LP model can raise the net return by 6.7%.

CONCLUSIONS

An LP model that establishes the ideal crop combination for the Rabi and Kharif seasons for a small farmer in the Gurugram district of Haryana is created in this study. Crops considered in the Rabi season are Wheat, Mustard, and Barley (Joo) whereas Crops considered in the Kharif season are Millet (Bajra), Rice, and Radish. For Rabi Season planting the LP model suggests planting more Mustard will increase the net returns by up to 24% and for the Kharif season planting more Rice on available land can increase net returns by up to 6.7%.

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Table 1: Farmer's methodology

Crops	Land(acres)	Man-days (days/acres)	Operating Capital (rupees/ acres)	Output (rupees/ acres)
Wheat	22	11	18000	50000
Mustard	13	4	14000	52000
Barley	5	11	15000	48000
Total	40	500	750000	

The net return = 1666000

Table 2: The Cropping Schedule as recommended by the LP Model

Crops	Land(acre)
Wheat	3
Mustard	35
Barley	2
Total	40

The net return = 2066000

Table 3: Comparison of Farmer's allocation and LP model allocation

Crops	Land(acres) allocation by Farmer	Land(acres) allocation by LP model
Wheat	22	3
Mustard	13	35
Barley	5	2
Total	40	40

Table 4:Farmer's methodology

Crops	Land(acres)	Man-days (days/acres)	Operating capital (rupees/ acres)	Output (rupees/ acres)
Millet (Bajra)	15	15	15000	35000
Rice	7	25	28000	70400

Radish	10	40	33400	85000
Total	32	800	950000	

The net return = 1867000

Table 5: The cropping schedule as recommended by the LP model

Crops	Land(acres)
Millet	10
Rice	15.34
Radish	6.67
Total	32

The Net Return =1996000

Table 6: Comparison of Farmer's allocation and LP model allocation

Crops	Land (acres) allocation by Farmer	Land (acres) allocation by LP Model
Millet (Bajra)	15	10
Rice	7	15.34
Radish	10	6.67
Total	32	32

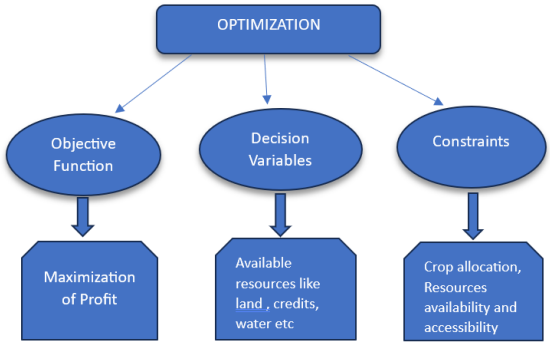


Figure 1: Optimization Model

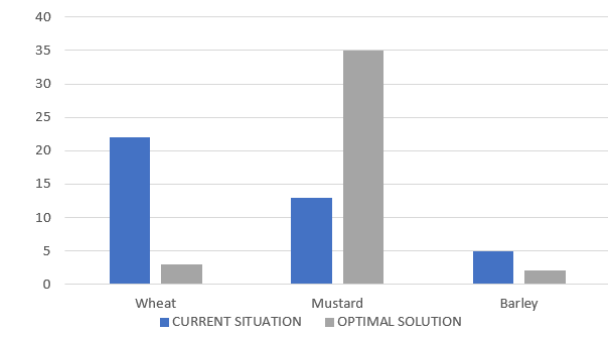


Figure 2: Comparison of Land allocation

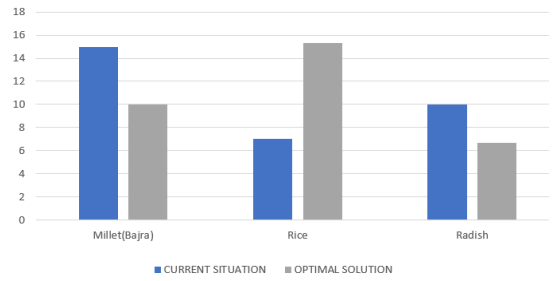


Figure 3: Comparison of land allocation

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